

The Conjugates of Related Primitive Words *

Chen-Ming Fan

Department of Information System
National Chin-Yi University of Technology, Taichung, Taiwan 411

C. C. Huang †

Department of Information Management Science
Chung Shan Medical University, Taichung, Taiwan 402

Abstract

This paper concerns the prefix-primitive (shortly, p-primitive) conjugates of related primitive words. The non-p-primitive conjugates of square-free words are studied. We show that every d-primitive square-free word over an alphabet with three elements has at most one non-p-primitive conjugate. We also find that all the conjugates of a primitive word are nor p-primitive.

1 Introduction

Beside primitive words and d-primitive words, no beginning square words are introduced and studied in [3]. Since every no beginning square word is primitive, no beginning square words are called prefix primitive words, shortly *p-primitive words*. A word is a *conjugate* of another word v if it is a cyclic permutation of v . Moreover, let the word u is a cyclic permutation of a word v . If u is a p-primitive word, then u is p-primitive conjugate of v . It is known that every conjugate of a primitive word is primitive [7]. However conjugates of primitive words are not all p-primitive. For instance, aab is not a p-primitive word whereas it is a conjugate of primitive word aba . This study motivates the investigation concerning p-primitive conjugates of related primitive words. Does that every primitive word exist a p-primitive conjugate?

A non-empty word which contains no subword of the form u^2 , where u is a non-empty word, is called *square-free*. For properties concerning square-free words, one is referred to [4]. Note that every square-free word is p-primitive. A word u is a *cyclic square-free word* if and only if every

conjugate of u is square-free [1]. Clearly, a word u is cyclic square-free if and only if every conjugate of u is p-primitive. Non-overlapping words are also called d-primitive words (see [2]). Every cyclic square-free word is d-primitive. But there are d-primitive square-free words which are not cyclic square-free. For instance, $ababc$ is not a square-free word whereas it is a conjugate of d-primitive square-free word $babca$. Thus a d-primitive square-free word could have many non-p-primitive conjugates. We shall show that a d-primitive square-free word over an alphabet with three elements has at most one non-p-primitive conjugate. Moreover, there exists a primitive word which the conjugates of this primitive word are not p-primitive.

2 Preliminaries

Let X be a finite alphabet and X^* the free monoid generated by X . Any element of X^* is called a *word*. The length of w is denoted by $\lg(w)$. Any subset of X^* is called a *language* over X . Let $X^+ = X^* \setminus \{1\}$, where 1 is the empty word. A word $w \in X^+$ is said to be *primitive* if $w = u^n$ with $u \in X^+$ implies $n = 1$. Let Q denote the set of all primitive words. For a language L , let $L^{(i)} = \{w^i \mid w \in L\}$ for any $i \geq 1$. The partial order relation $\leq_p$ (resp. $\leq_s$) is defined as: for $u, v \in X^*$, $u \leq_p v$ (resp. $u \leq_s v$) if and only if $v \in uX^*$ (resp. $v \in X^*u$). A word u is said to be a *prefix* (resp. *suffix*) of another word w if $u \leq_p w$ (resp. $v \leq_s w$). For a given word $x \in X^+$, we define the infix of the word x as $E(x) = \{y \in X^+ \mid x \in X^*yX^*\}$. A word w has a *prefix n-power* if $w \in u^nX^*$ for some $u \in X^+$. For $w \in X^+$, let $N(w)$ denote the maximal integer n such that w has a prefix n -power. For any $i \geq 1$,

*AMS Subject Classification: 68R15

†Corresponding author E-mail: cchuang@csmu.edu.tw

we define P_i as $P_i = \{w \in X^+ \mid N(w) = i\}$. From the above definition, it is clear that $P_i \cap P_j = \emptyset$ for every $i \neq j$ and $X^+ = \bigcup_{i \geq 1} P_i$. Every word w in P_1 is said to be *prefix primitive* (shortly, *p-primitive*), i.e., $w \notin u^2 X^*$ for any $u \in X^+$. For example: let $X = \{a, b\}$. Then ab^n is a p-primitive word for any $n \geq 1$. A word u is a *conjugate* of another word v if there is a word w such that $uw = wv$. The partial order relation $\leq_d$ is defined as: for $u, v \in X^*$, $u \leq_d v$ if and only if $v = ux = yu$ for some $x, y \in X^*$. Note that $1 \leq_d v$ for any $v \in X^+$. A word $u \in X^+$ is said to be *d-primitive* if for any $v \in X^+$, $v \leq_d u$ implies $u = v$. Let $D(1)$ be the set of all d-primitive words in X^* (see [2]). It is clear that $D(1)$ is a subset of Q . For a word $u \in X^+$, let $\text{Pref}(u) = \{v \in X^+ \mid v \leq_p u\}$ and $\text{Suff}(u) = \{v \in X^+ \mid v \leq_s u\}$. The following lemmata are well known properties of conjugates, which will be needed in the sequel.

Lemma 2.1 ([5]) *If $uw = wv$ for $u, v, w \in X^*$ with $u \neq 1$, then $u = xy$, $w = (xy)^n x$, $v = yx$ for some $x, y \in X^*$ and $n \geq 0$.*

By virtue of Lemma 2.1, a word u is a conjugate of another word v iff $u = xy$ and $v = yx$ for some $x, y \in X^*$. Hence a conjugate word of a word u is also called a *cyclic permutation* of u . Let $\sigma(u) = \{v \mid v \text{ is a conjugate of } u\}$.

Lemma 2.2 ([7]) *Let $uv = x^i$, $u, v \in X^+$, $x \in Q$, $i \geq 1$. Then $vu = y^i$ for some $y \in Q$.*

Moreover, we quote the following lemmata concerning basic properties of catenations and powers of words.

Lemma 2.3 ([5]) *Let $u, v \in X^+$. Then $uv = vu$ implies that u and v are powers of a common word.*

Lemma 2.4 ([5]) *If $u^m = v^n$ and $m, n \geq 1$, then u and v are powers of a common word.*

3 The Non-P-Primitive Conjugates of Related Primitive Words

In this section, we will study p-primitivities of the conjugates of square-free words. It is known that every cyclic square-free word is d-primitive and square-free [1]. However, there are d-primitive square-free words which are not cyclic square-free. For example: Let $X = \{a, b, c, d, e, f\}$ and $u = abeaceabeacfdbeadb$. Then u is a d-primitive square-free word with $|\sigma(u) \setminus P_1| =$

3, that is, $(eabeac)^2 fdbeadb, (bea)^2 ceabeacfdbeadb$, and $(dbea)^2 beaceabeacf \notin P_1$. This observation shows that a d-primitive square-free word could have many non-p-primitive conjugates. First, we derive that a algorithm to construct non-p-primitive words which are the conjugate of d-primitive square-free word.

Lemma 3.1 ([6]) *Let $u \in X^+ \setminus D(1)$. Then there exists $v \in X^+$ with $\lg(v) \leq \lg(u)/2$ such that $v <_d u$.*

Lemma 3.2 *Let $u, v \in X^+$. Then $uv \notin D(1)$ if and only if one of the following conditions holds:*

- (i) *There exists $x \in X^+$ such that x is a prefix of u and a suffix of v .*
- (ii) *There exist $x \in X^+$, $y \in X^*$ such that $u = xvyx$.*
- (iii) *There exist $x \in X^+$, $y \in X^*$ such that $v = xyux$.*

Proof. ($\Leftarrow$) Immediate.

($\Rightarrow$) Suppose that $uv \notin D(1)$. Then by Lemma 3.1, there exist $r \in X^+, s \in X^*$ such that $uv = rsr$. We consider the following cases:

(1) $\lg(u) \geq \lg(v)$. If $\lg(r) \leq \lg(v)$, then there exists $x \in X^+$ such that $x = r$ which is a prefix of u and a suffix of v ; hence (i) holds. If $\lg(r) > \lg(v)$, then there exists $x \in X^+$ such that $r = xv$. This in conjunction with $uv = rsr$ yields that $u = xvsx$; hence (ii) holds.

(2) $\lg(u) < \lg(v)$. If $\lg(r) \leq \lg(u)$, then there exists $x \in X^+$ such that $x = r$ which is a prefix of u and a suffix of v ; hence (i) holds. If $\lg(r) > \lg(u)$, then there exists $x \in X^+$ such that $r = ux$. This in conjunction with $uv = rsr$ yields that $v = xyux$; hence (iii) holds.

Proposition 3.1 *Let u be a d-primitive square-free word. If there exist a conjugate of u is not p-primitive, then $u = xyxzy$ for some $x, y, z \in X^+$ with $xy \in D(1)$, $x \not\leq_s xz$ or $u = xzyxy$ for some $x, y, z \in X^+$ with $xy \in D(1)$, $y \not\leq_p zy$.*

Proof. Let u be a d-primitive square-free word such that there is a conjugate of u which is not p-primitive. That is, there exist $u_1, u_2 \in X^+$ such that $u = u_1 u_2$ and $u_2 u_1 = v_1^2 v_2$ for some $v_1 \in X^+$ and $v_2 \in X^*$. Since u is square-free which is also primitive word, by virtue of Lemma 2.2, $v_2 \in X^+$. In the meanwhile, the square-freeness of u also implies that $\lg(u_2) < \lg(v_1^2)$. Moreover, $u_2 \neq v_1$ because u is a d-primitive word. As $\lg(u_2) < \lg(v_1)$, we have that $v_1 = u_2 v_3$

and $u = v_3u_2v_3v_2u_2$ for some $v_3 \in X^+$. Let $v_3 = x$, $u_2 = y$ and $v_2 = z$. Then $u = xyxzy$ where $x, y, z \in X^+$. As $\lg(u_2) > \lg(v_1)$, we have that $v_1 = v_3v_4$, $u_2 = v_1v_3$, and $u_1 = v_4v_2$ for some $v_3, v_4 \in X^+$. Let $v_4 = x$, $v_2 = z$ and $v_3 = y$. Then $u = xzyxy$. Since u is d-primitive, $\text{Pref}(x) \cap \text{Suff}(y) = \emptyset$. There are the following csaes:

(1) $u = xyxzy$. If $x \leq_s xz$, then there exists $x' \in X^+$ such that $xz = x'x$. This yields that $u = xyx'xy \notin D(1)$, a contradiction; hence $x \not\leq_s xz$. Now we suppose that $xy \notin D(1)$. By Lemma 3.2, one of the following conditions holds:

(i) There exists $r \in X^+$ such that r is a prefix of x and a suffix of y .

(ii) There exist $r \in X^+$, $s \in X^*$ such that $x = rysr$.

(iii) There exist $r \in X^+$, $s \in X^*$ such that $y = rsxr$. If (i) holds, then it contradicts that $\text{Pref}(x) \cap \text{Suff}(y) = \emptyset$. If (ii) holds, then $u = rysryrysryz = rys(ry)^2srzy$ is not square-free, a contradiction. If (iii) holds, then $u = xrsxrzsxr \notin D(1)$, a contradiction; hence $xy \in D(1)$.

(2) $u = xzyxy$. If $y \leq_p zy$, then there exists $y' \in X^+$ such that $yy' = zy$ such that $u = xyy'xy \notin D(1)$, a contradiction; hence $y \not\leq_p zy$. Now we suppose that $xy \notin D(1)$. It is Similar to case (1). If (i) holds, then it contradicts that $\text{Pref}(x) \cap \text{Suff}(y) = \emptyset$. If (ii) holds, then $u = rysrzsryrsry \notin D(1)$, a contradiction. If (iii) holds, then $u = xzrsxrzsxr = xzrs(xr)^2rsxr$ is not square-free, a contradiction; hence $xy \in D(1)$.

In the following, we will construct d-primitive square-free words. We use method, which has made up square-free words provided on [1], to construct d-primitive square-free words.

Let $w_{10} = a$, $w_{20} = b$, $w_{30} = c$, $w_{40} = c$. One can define as follow:

$$w_{1(n+1)} = w_{1n}w_{2n}w_{4n}w_{2n}w_{1n}w_{3n}w_{2n}w_{3n}w_{1n}w_{4n}w_{2n}w_{2n}w_{1n}w_{3n}w_{1n}w_{2n}w_{1n}w_{4n}w_{2n},$$

$$w_{2(n+1)} = w_{1n}w_{2n}w_{4n}w_{2n}w_{1n}w_{3n}w_{2n}w_{3n}w_{1n}w_{2n}w_{1n}w_{3n}w_{1n}w_{2n}w_{3n}w_{1n}w_{4n}w_{2n},$$

$$w_{3(n+1)} = w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}w_{1n}w_{2n}w_{3n}w_{1n}w_{2n}w_{1n}w_{3n}w_{1n}w_{2n}w_{3n}w_{1n}w_{4n}w_{2n},$$

$$w_{4(n+1)} = w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}w_{1n}w_{2n}w_{3n}w_{1n}w_{4n}w_{2n}w_{1n}w_{3n}w_{1n}w_{2n}w_{1n}w_{4n}w_{2n}.$$

For $n = 0$, we have $w_{11} = abcbacbcacbacabcb$,

$$w_{21} = abcbacbcabacabcb,$$

$$w_{31} = cabcbacabacabcb,$$

$$w_{41} = cabcbacbcacabcb.$$

Lemma 3.3 ([1]) *The words $w_{1n}, w_{2n}, w_{3n}, w_{4n}, w_{1n}w_{2n}, w_{1n}w_{3n}, w_{1n}w_{4n}, w_{2n}w_{3n}, w_{2n}w_{4n}$ are cyclic-square-free and $\text{Pref}(w_{in}) \cap \text{Suff}(w_{jn}) = \emptyset$ for every $n \geq 1$ and $1 \leq i, j \leq 4$.*

Lemma 3.4 *The words $w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}, w_{3n}w_{1n}w_{2n}w_{3n}w_{1n}, w_{1n}w_{2n}w_{3n}w_{1n}w_{2n},$ and $w_{4n}w_{2n}w_{3n}w_{1n}w_{2n}$ are all square-free for any $n \geq 1$.*

Proof. By Lemma 3.3, the word $w_{1(n+1)}w_{3(n+1)}$ is cyclic-square-free. From the definition of cyclic-square-free, the subword $w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}, w_{3n}w_{1n}w_{2n}w_{3n}w_{1n}, w_{1n}w_{2n}w_{3n}w_{1n}w_{2n},$ and $w_{4n}w_{2n}w_{3n}w_{1n}w_{2n}$ of $w_{1(n+1)}w_{3(n+1)}$ are all square-free for any $n \geq 1$.

Proposition 3.2 *The word $w_{1n}w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}w_{3n}$ is d-primitive square-free for every $n \geq 1$.*

Proof. For convenience, let $u_n = w_{1n}w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}w_{3n}$ for any $n \geq 1$. First, we show that u_n is d-primitive. Assume that u_n is not d-primitive. Then by Lemma 3.1, there exists $v \in X^+$ with $\lg(v) \leq \lg(u_n)/2$ such that $u_n = vsv$ for some $s \in X^*$. Since $\lg(w_{in}) = \lg(w_{jn})$ for every $n \geq 1$ and $1 \leq i, j \leq 4$, we let $m = \lg(w_{in})$ for any $i = 1, 2, 3, 4$. Then $\lg(u_n) = 8m$ and $\lg(v) \leq 4m$. Note that $\text{Pref}(w_{1n}) \cap \text{Suff}(w_{3n}) = \emptyset$. This in conjunction with $w_{1n}w_{2n} \leq_p u_n$ and $w_{4n}w_{2n}w_{3n} \leq_s u_n$ yields that $\lg(v) \geq 2m$. It is clear that $\lg(v) \neq km$ for any $2 \leq k \leq 4$. Indeed, if $\lg(v) = km$ for any $2 \leq k \leq 4$, then $w_{1n} = w_{in}$ for some $i = 2, 3, 4$, a contradiction. We now consider the following cases:

(1) $2m < \lg(v) < 3m$. There exists a suffix t of v such that t is a suffix of w_{3n} is also a prefix of w_{3n} . That is, $t \in \text{Pref}(w_{3n}) \cap \text{Suff}(w_{3n})$, a contradiction.

(2) $3m < \lg(v) < 4m$. There exists a suffix t of v such that t is a suffix of w_{3n} is also a prefix of w_{1n} which contradicts that $t \in \text{Pref}(w_{1n}) \cap \text{Suff}(w_{3n})$, a contradiction.

Next, we show that u_n is square-free. Assume that u_n is not square-free. Then there exists $w \in X^+$ such that $w^2 \in E(u_n)$. Since $w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}$ is square-free, by Lemma 3.4, we consider the following cases:

(1) $w^2 \in E(w_{1n}w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n})$. Since $w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}$ is square-free, there exists $x \in X^+$ with $x \leq_s w_{1n}$ such that $w^2 \leq_p xw_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}$. Note that $x <_p w$. Indeed, if $w \leq_p x$, then $\lg(w) \leq \lg(x)$. This

in conjunction with $x \leq_s w_{1n}$ and $\lg(w_{1n}) = \lg(w_{2n})$ yields that $w^2 \in E(w_{1n}w_{2n})$, which contradicts that $w_{1n}w_{2n}$ is square-free. From $w_{1n}w_{2n}w_{3n}w_{1n}w_{2n}$ is square-free by Lemma 3.4, we need only to consider the following two subcases:

(1-1) $w^2 = xw_{2n}w_{3n}w_{1n}w_{2n}y$ for some $y \in X^+$ with $y \leq_p w_{4n}$. There exist $z_1, z_2 \in X^*$ such that either (i) $w_{3n} = z_1z_2$, $z_1 \neq 1$ and $w = xw_{2n}z_1 = z_2w_{1n}w_{2n}y$, or (ii) $w_{1n} = z_1z_2$ and $w = xw_{2n}w_{3n}z_1 = z_2w_{2n}y$. Assume that (i) holds. Then $\lg(xw_{2n}z_1) = \lg(z_2w_{1n}w_{2n}y)$. This implies that $\lg(x) + \lg(z_1) = \lg(z_2) + \lg(z_1z_2) + \lg(y)$; hence $\lg(x) > \lg(z_2)$. There exists $x' \in X^+$ such that $x = z_2x'$ and $x'w_{2n}z_1 = w_{1n}w_{2n}y$; hence $x' <_p w_{1n}$. This in conjunction with $x' <_s x \leq_s w_{1n}$ yields that $x' \in \text{Pref}(w_{1n}) \cap \text{Suff}(w_{1n})$, a contradiction. Moreover, assume that (ii) holds. Then $\lg(xw_{2n}w_{3n}z_1) = \lg(z_2w_{2n}y)$. This implies that $\lg(x) + \lg(z_1z_2) + \lg(z_1) = \lg(z_2) + \lg(y)$; hence $\lg(y) > \lg(z_1)$. There exists $z' \in X^+$ such that $y = z'z_1$ and $xw_{2n}w_{3n} = z_2w_{2n}z'$; hence $z' <_s w_{3n}$. This in conjunction with $z' <_p y \leq_p w_{4n}$ yields that $z' \in \text{Pref}(w_{4n}) \cap \text{Suff}(w_{3n})$, a contradiction.

(1-2) $w^2 = xw_{2n}w_{3n}w_{1n}w_{2n}w_{4n}y$ for some $y \in X^+$ with $y \leq_p w_{2n}$. There exist $z_1, z_2 \in X^+$ such that $w_{1n} = z_1z_2$ and $w = xw_{2n}w_{3n}z_1 = z_2w_{2n}w_{4n}y$. If $\lg(x) \geq \lg(z_2)$, then there exists $x' \in X^*$ such that $x = z_2x'$ and $x'w_{2n}w_{3n}z_1 = w_{2n}w_{4n}y$. As $x' = 1$. This implies that $w_{3n} = w_{4n}$, a contradiction; hence $x' <_p w_{2n}$ where $x' \in X^+$. This in conjunction with $x' <_s x \leq_s w_{1n}$ yields that $x' \in \text{Pref}(w_{2n}) \cap \text{Suff}(w_{1n})$, a contradiction. If $\lg(x) < \lg(z_2)$, then there exists $z' \in X^+$ such that $z_2 = xz'$ and $w_{2n}w_{3n}z_1 = z'w_{2n}w_{4n}y$; hence $z' <_p w_{2n}$. This in conjunction with $z' <_s z_2 <_s w_{1n}$ yields that $z' \in \text{Pref}(w_{2n}) \cap \text{Suff}(w_{1n})$, a contradiction.

(2) $w^2 \in E(w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}w_{3n})$. Since $w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}$ is square-free, there exists $y \in X^+$ with $y \leq_p w_{3n}$ such that $w^2 \leq_s w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}y$. Note that $y <_s w$. Indeed, if $w \leq_s y$, then $\lg(w) \leq \lg(y)$. This in conjunction with $y \leq_p w_{3n}$ and $\lg(w_{3n}) = \lg(w_{2n})$ yields that $w^2 \in E(w_{2n}w_{3n})$, which contradicts that $w_{2n}w_{3n}$ is square-free. From $w_{4n}w_{2n}w_{3n}$ is square-free by Lemma 3.4, we need to consider the following four subcases:

(2-1) $w^2 = xw_{4n}w_{2n}y$ for some $x \in X^+$ with $x \leq_s w_{2n}$. There exist $z_1, z_2 \in X^+$ such that either (i) $w_{4n} = z_1z_2$ and $w = xz_1 = z_2w_{2n}y$, or (ii) $w_{2n} = z_1z_2$ and $w = xw_{4n}z_1 = z_2y$. As-

sume that (i) holds. Then $\lg(xz_1) = \lg(z_2w_{2n}y) \leq \lg(w_{2n}z_1)$ because $x \leq_s w_{2n}$. This implies that $\lg(z_1) \geq \lg(z_2y)$; hence $\lg(z_1) > \lg(y)$. There exists $z' \in X^+$ such that $z_1 = z'y$ and $xz' = z_2w_{2n}$; hence $z' <_s w_{2n}$. This in conjunction with $z' <_p z_1 \leq_p w_{4n}$ yields that $z' \in \text{Pref}(w_{4n}) \cap \text{Suff}(w_{2n})$, a contradiction. Moreover, assume that (ii) holds. Then $\lg(xw_{4n}z_1) = \lg(z_2y)$. This implies that $\lg(x) + \lg(z_1z_2) + \lg(z_1) = \lg(z_2) + \lg(y)$; hence $\lg(y) > \lg(z_1)$. There exists $y' \in X^+$ such that $y = y'z_1$ and $xw_{4n} = z_2y'$; hence $y' <_s w_{4n}$. This in conjunction with $y' <_p y \leq_p w_{3n}$ yields that $y' \in \text{Pref}(w_{3n}) \cap \text{Suff}(w_{4n})$, a contradiction.

(2-2) $w^2 = xw_{2n}w_{4n}w_{2n}y$ for some $x \in X^+$ with $x \leq_s w_{1n}$. There exist $z_1, z_2 \in X^+$ such that $w_{4n} = z_1z_2$ and $w = xw_{2n}z_1 = z_2w_{2n}y$. If $\lg(y) > \lg(z_1)$, then there exists $y' \in X^+$ such that $y = y'z_1$ and $xw_{2n} = z_2w_{2n}y'$; hence $y' <_s w_{2n}$. This in conjunction with $y' <_p y \leq_p w_{3n}$ yields that $y' \in \text{Pref}(w_{3n}) \cap \text{Suff}(w_{2n})$, a contradiction. If $\lg(z_1) > \lg(y)$, then there exists $z' \in X^+$ such that $z_1 = z'y$ and $xw_{2n}z' = z_2w_{2n}$; hence $z' <_s w_{2n}$. This in conjunction with $z' <_p z_1 \leq_p w_{4n}$ yields that $z' \in \text{Pref}(w_{4n}) \cap \text{Suff}(w_{2n})$, a contradiction.

(2-3) $w^2 = xw_{1n}w_{2n}w_{4n}w_{2n}y$ for some $x \in X^+$ with $x \leq_s w_{3n}$. This subcase is similar to subcase (2-1).

(2-4) $w^2 = xw_{3n}w_{1n}w_{2n}w_{4n}w_{2n}y$ for some $x \in X^+$ with $x \leq_s w_{2n}$. This subcase is similar to subcase (2-2).

(3) $w^2 = xw_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}y$ for some $x, y \in X^+$ with $x \leq_s w_{1n}$, $y \leq_p w_{3n}$. There exist $z_1, z_2 \in X^+$ such that either (i) $w_{1n} = z_1z_2$ and $w = xw_{2n}w_{3n}z_1 = z_2w_{2n}w_{4n}w_{2n}y$, or (ii) $w_{2n} = z_1z_2$ and $w = xw_{2n}w_{3n}w_{1n}z_1 = z_2w_{4n}w_{2n}y$. These subcases are similar to subcase (3-1).

Proposition 3.3 *Let $|X| \geq 3$. Then for any $n \geq 1$, there exists a d -primitive square-free word u such that $|\sigma(u) \setminus P_1| \geq n$.*

Proof. Let $n \geq 1$ and let $u = w_{1n}w_{2n}w_{3n}w_{1n}w_{2n}w_{4n}w_{2n}w_{3n}$. Then by Proposition 3.2, u is a d -primitive square-free word. Let $x = w_{1n}$, $y = w_{2n}$, $z = w_{3n}$, $v = w_{4n}$. We rewrite $u = xyzxyvyz$. Since $\lg(y) = 18^n > n$, there exist $y_1, y_2, \dots, y_n \in X^+$ such that $y = y_1y_2 \cdots y_n$; hence $u = xy_1y_2 \cdots y_nzxy_1y_2 \cdots y_nv y_1y_2 \cdots y_nz$. Let $p_i = xy_1y_2 \cdots y_nzxy_1y_2 \cdots y_nv y_1 \cdots y_i$, $q_i = y_{i+1} \cdots y_nz$ for $i = 1, 2, \dots, n-1$ and $p_n = xy_1y_2 \cdots y_nzxy_1y_2 \cdots y_nv y_1 \cdots y_n$, $q_n = z$. Then $u = p_1q_1 = \cdots = p_nq_n$ and $q_i p_i = (y_{i+1} \cdots y_nz)(xy_1y_2 \cdots y_nzxy_1y_2 \cdots y_nv y_1 \cdots y_i) =$

$(y_{i+1} \cdots y_n z x y_1 \cdots y_i)^2 y_{i+1} \cdots y_n v y_1 \cdots y_i$
 for $i = 1, 2, \dots, n-1$, $q_n p_n =$
 $z(x y_1 y_2 \cdots y_n z x y_1 y_2 \cdots y_n v y_1 \cdots y_n) =$
 $(z x y_1 y_2 \cdots y_n)^2 v y_1 \cdots y_n$. All of $q_i p_i$, $i = 1, 2, \dots, n$
 are not p-primitive. Therefore $|\sigma(u) \setminus P_1| \geq n$.

Let $a \neq b$ be two letters and let $\sigma(u)$ be the set containing all conjugates of u . If $u = aba$, then $\sigma(u) = \{aba, aab, baa\}$. There exists a conjugate of a primitive word aba which is p-primitive. Note that the conjugate of the primitive word is primitive. From the definition of p-primitive word, the p-primitive word is primitive. Does the conjugate of a primitive word exist p-primitive word? We have the following proposition.

Proposition 3.4 *Let $|X| \geq 2$. If $(x^2yxy)^2x^2y \in Q$ for some $x, y \in X^+$ with $\sqrt{x} \neq \sqrt{y}$, then every conjugate of $(x^2yxy)^2x^2y$ is not p -primitive.*

Proof. Let $u = (x^2yxy)^2x^2y \in Q$ for some $x, y \in X^+$ with $\sqrt{x} \neq \sqrt{y}$. Suppose that there is a word $u_i \in \sigma(u)$ for some i such that u_i is a p-primitive word. Then we consider the following subcases: (A) $u_1 = (xxyxyxyxyxy) = x^2yxyxyxyxy \not\in P_1$, (B) $u_2 = (xyxyxyxyxy)(x) = (xy)^2xyxyxyxy \not\in P_1$, (C) $u_3 = (yxyxyxyxy)(xx) = (yx)^2xyxyxyxx \not\in P_1$, (D) $u_4 = (xyxyxyxy)(xy) = (yx)^2yxyxyxy \not\in P_1$, (E) $u_5 = (yxyxyxy)(xyx) = (yxx)^2yxy \not\in P_1$, (F) $u_6 = (xxyxyxy)(xyxy) = x^2yxyxyxyxy \not\in P_1$, (G) $u_7 = (xyxyxy)(xyxyx) = (xy)^2xyxyxyxy \not\in P_1$, (H) $u_8 = (yxyxy)(xyxyxx) = (yx)^2xyxyxyxx \not\in P_1$, (I) $u_9 = (xyxy)(xyxyxy) = (yx)^2xyxyxy \not\in P_1$, (J) $u_{10} = (yxy)(xyxyxyxy) = (yxx)^2yxyxyxy \not\in P_1$, (K) $u_{11} = (xxy)(xyxyxyxy) = (xy)^2xyxyxy \not\in P_1$, (L) $u_{12} = (xy)(xyxyxyxyx) = (xyx)^2yxyxyxy \not\in P_1$, (M) $u_{13} = (y)(xyxyxyxyxx) = (yxx)^2yxx \not\in P_1$.

Therefore, every conjugate of $(x^2yxy)^2x^2y$ is not p-primitive.

We give one example from the above proposition. For instance, word $aababaababab$ is primitive. The set $\sigma(aababaababab) = \{a^2babaababab, (ab)^2aababaab, (ba)^2ababaab, (aba)^2baabab, (baaba)^2aba, a^2babaabaabab, (ab)^2aabaababa, (ba)^2abaababaa, (aba)^2ababaab, (baa)^2babaaba, (aab)^2abaabab, (aba)^2baababa, (baaba)^2baa\} \not\subset P_1$.

References

- [1] C. M. Fan, H. J. Shyr and S. S. Yu, "d-Words and d-Languages", *Acta Informtica*, Vol.**35**, pp. 709–727, 1998.
- [2] S. C. Hsu, M. Ito and H. J. Shyr, "Some Properties of Overlapping Order and Related Languages", *Soochow Journal of Mathematics*, Vol.**15**, pp. 29–45, 1989.
- [3] C. C. Huang, S. S. Yu and Yu-Kuang Zhao, "Prefix Primitive Languages", *Journal of Information & Optimization Sciences*, Vol.**25**, No.**2**, pp. 303–316, 2004.
- [4] M. Lothaire, *Combinatorics on Words*, Cambridge University Press, United Kingdom, 1983.
- [5] R. C. Lyndon and M. P. Schützenberger, "On the Equation $a^M = b^N c^P$ in a Free Group", *Michigan Mathematical Journal*, Vol.**9**, pp. 289–298, 1962.
- [6] H. J. Shyr, *Free Monoids and Languages*, Third Edition, Hon Min Book Company, Taichung, Taiwan, 2001.
- [7] H. J. Shyr and G. Thierrin, "Disjunctive Languages and Codes", *Fundamentals of Computation Theory, Proceedings of the 1977 Inter. FCT-Conference*, Poznan, Poland, Lecture Notes in Computer Science, No.**5**, Springer-Verlag, pp. 171–176, 1977.